COURSE SYLLABUS

Computer Science Math

2026–27 · The Cushman School

TEACHER
Mr. Willie Avendano

EMAIL
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ROOM
Room 502/3

CLASS MEETS
Monday, Period 3 (9:41–10:21) · Tuesday & Thursday, Block 3 (10:59–12:19)

EXTRA HELP
3:15–4:00 PM on Mondays, Tuesdays, and Wednesdays, or by appointment. *Dismissal runs 3:00–3:15, so help begins at 3:15. Thursdays are unavailable — HS faculty meeting.*

Course Description

A project-based mathematics course. Students build working mathematical models — statistics, simulation, forecasting, and optimization — primarily in **Excel**, and every unit is organized around a cornerstone project: a model students design and build to demonstrate the skills they have practiced, then present to the class.

The spreadsheet is the lab; the projects are the point. Each cornerstone feeds the next, so skills compound across the year the way the models do. No prior experience is required.

Units

Unit 0 The Toolbox

- Notation (Σ, Δ, functions) and estimation
- Excel training wheels — cells, formulas, charts
- Lookups and the KDEA protocol

Unit 1 Cornerstone I — Food & Macro Tracker

- Structuring data
- VLOOKUP tables and SUMIF aggregation
- Honest charts

`Unit 2` **Cornerstone II — Financing a Car**

- Compound interest
- Amortization and payment formulas
- Scenario comparison

`Unit 3` **Cornerstone III — Personal Budget**

- Income versus spending
- Fixed and variable costs
- Savings rate and scenario planning

`Unit L` **The Casino Lab**

- Probability and expected value
- RAND simulations and the law of large numbers
- The house edge

`Unit 4` **Cornerstone IV — Stocks: Forecasting & Management**

- Time series and moving averages
- Volatility and risk
- Monte Carlo simulation

`Unit 5` **Cornerstone V — The Lemonade Stand**

- Simulated demand and pricing
- Costs and break-even
- Sensitivity analysis

`Unit C` **Capstone — The Pop-Up Forecast**

- Real Design District data
- Regression and seasonality
- Solver optimization and the pitch

Week-by-week pacing — including which weeks carry no new content — is on the course page.

Required Texts and Links

Subject to change and grow.

- Course-issued starter workbooks
- Microsoft Excel (Office 365 for Education)
- Optional: Python in Excel for the power lane

Course Materials

- School Laptop
- Internet Connection
- Computer Charger
- Notebook, pencil, pen

Grades and Evaluations

Grades will be calculated using the 4.0 scale as mentioned in the student handbook.

70%

Summative

cornerstone projects, their presentations, and the year-end capstone pitch

25%

Formative

in-class model building, guided spreadsheet exercises, skills checks, class discussions, and unannounced quizzes

5%

Diagnostics

Entrance and exit tickets, and other checks that are neither summative nor formative.

Description of Common Assessments

Cornerstone projects are the summative assessments. Each is a working model you design and build, then present to the class. Rubrics are distributed when the cornerstone is introduced and used to score the finished workbook and the presentation.

The workbook is the work — models are graded on whether they function, whether the formulas are honest, and whether someone else could read them. Hard-coded numbers where a formula belongs will cost you credit.

The optional Python power lane opens during the Monte Carlo and capstone weeks for students ready to push past native formulas. Excel remains the shared floor; the power lane is enrichment, not a separate track.

Classwork (CW in Veracross) should be completed during the allotted class time and is **graded for accuracy**. Since these assessments are formative, students will have the chance to resubmit work within an allotted time determined by the teacher.

Homework (HW in Veracross) assignments allow students to practice their skills outside of the school setting or prepare themselves for the following class. Homework assignments are due at the start of the class period.

Scoring on homework will be as follows:

- **4** — All components of the assignment are completed in full, addressing all requirements with a high level of detail.

- **3** — Most components of the assignment are completed, meeting most of the requirements with satisfactory detail.
- **2** — Some components of the assignment are completed, but gaps or missing elements are noticeable. Detail is lacking in some areas.
- **1** — Few components of the assignment are completed, and significant gaps or missing elements are evident. Details are minimal.
- **NTI** — No assignment is turned in and is subject to the late work policy.

Rules and Routines

Your success in this course will directly reflect on the amount of time and energy you put into it.

Rules

Respect. This is the number one rule in the class. Respect you, your peers, and Mr. Avendano.

Consequences. Most violations will undergo the following consequences:

1. **1st Offense:** Verbal warning
2. **2nd Offense:** Conference at the end of class
3. **3rd Offense:** Behavior report to administration

Routines

1. **Class preparation.** Students are expected to come to class with materials ready, including and not limited to, having their laptops charged for class, with an adequate computer charger prepared to charge their computers if need be. I will have limited materials such as pens, pencils, calculators, and paper available. However, students who are always relying on these and do not consistently have their own materials will receive scores of 1 or 2 in responsibility (IRC).
2. **Participation.** There are many ways to participate in class and I expect students to partake in some way. This can include adding to class discussions, asking questions about class materials, working collaboratively during group work, attending extra help, etc.
3. **Absences.** Consistent attendance is strongly recommended as it facilitates the learning process. It is the student's responsibility to inform me via email or in person when they are or will be absent. Students will have two class period days to make up work from absences (Student Handbook, p. 21). Any work handed in afterwards will be subject to the "Late Work Policy."
4. **Food & Drinks.** Students may not eat or drink in the classroom. Water is the only exception. If you need to eat, you may briefly step outside when I allow permission. No visits to the vending machines are allowed.
5. **Late Work.** Students will have one week to submit any late work for partial credit. Late formative assessments can receive up to a 2.0 in the gradebook. It is not fair to give the same credit to students who turn their work in on time as to those who turn their work in late. However, if a student has an *extenuating circumstance*, I am more than willing to make accommodations. See me BEFORE the due date if this applies to you! **Late work will NOT receive credit after one week.** After the week, the highest possible grade is a 0. In this case, the student will receive a Complete/No Credit or an NTI if never turned in.

6. **Resubmissions/Test Corrections.** If a student obtains a score of 2 or lower on a written exam, they may request to complete test corrections only by filling out the necessary Resubmit Form. These are done only during extra help and in my presence. Students must arrange a time with me to complete corrections within a given deadline and will earn 1/2 credit back on their missed points. The original work must have been submitted on time to resubmit. Students who want to improve an unannounced quiz score may retake a similar quiz (once). This policy only applies to the aforementioned assignments. It does not apply to regular assignments, announced quizzes, or projects.
7. **Electronics.** Cell phones must be stored in the pouch, put away in the backpack, and on silent. Students are expected to use their laptops solely for academic purposes. A student who does not comply with this expectation is not showing respect for themselves, their classmates, or their teacher and will be in violation of the classroom rule. Along with the application of consequences, the student will receive scores of 1 or 2 in industry, courtesy, and responsibility.
8. **Restroom.** If a student needs to use the restroom, they may raise their hand and point at the door. When given permission, students may go. Due to the fact that class is on the fifth floor and the restroom is only permitted for faculty on the fifth, students are permitted to go to another floor for the restroom. The time used for the restroom must be kept to a minimum and may be subject to be tracked. Excessive use of the restroom will result in students using their passes for restroom use and/or parents and administration may be informed.

Academic Integrity

Students are expected and required to be honest in the preparation of assignments, papers, projects, and all other assessments. Cheating (including plagiarism) in any form is not allowed and it hinders the learning process. If a student chooses to do so, the work will receive a 0 and will lead to further disciplinary consequences. If a student cheats off a classmate's work, both will receive the above-mentioned outcomes.

Disclaimer

The institution and teacher reserve the right to make modifications to this information throughout the academic school year, should it be necessary.

Acknowledgement of Receipt and Understanding

Please sign and submit as soon as possible, but **no later than midnight on Sunday, August 30, 2026**.

> **Midnight means midnight.** Anything submitted at 12:01 AM Monday or later is late. Submissions are checked Monday morning.

I have read and fully understand the academic and behavioral expectations outlined in Mr. Avendano's Computer Science Math syllabus. I agree to abide by these guidelines or will undergo the academic and/or disciplinary consequences.

Student Printed Name ______________________________

Student Signature ______________________________

Parent/Guardian Printed Name ______________________________

Parent/Guardian Signature ______________________________

Date ______________________________